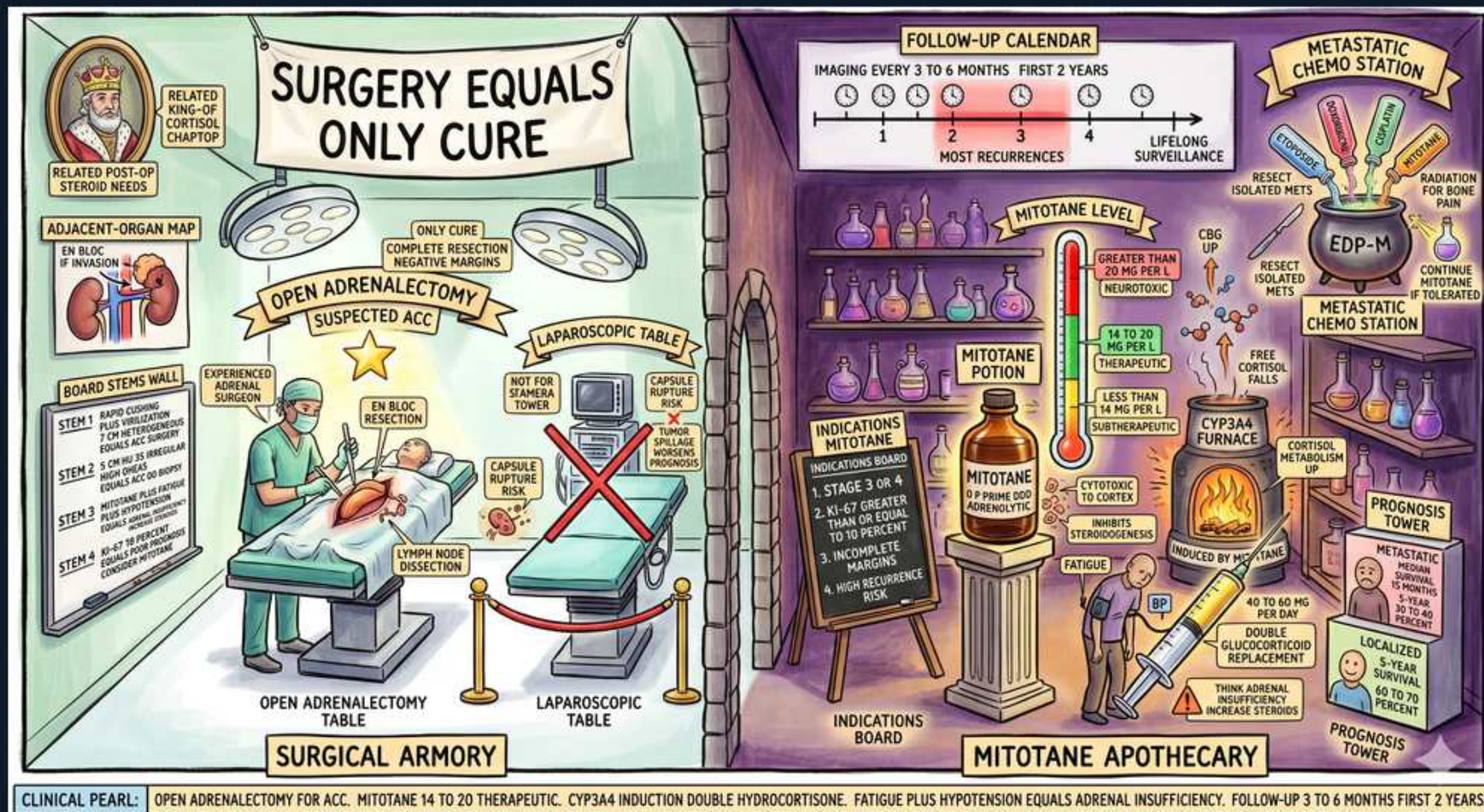


Adrenal — Surgical Armory & Mitotane Apothecary



1. Surgery = only cure

WHAT YOU SEE

Banner 'SURGERY EQUALS ONLY CURE'

WHAT IT ENCODES

Adrenocortical carcinoma (ACC) is a rare, aggressive tumor with a bimodal age distribution (young children and adults 40–50). Complete surgical RESECTION with negative (R0) margins is the ONLY potentially curative therapy; unresectable or metastatic disease has a poor prognosis. Functional tumors most often hypersecrete cortisol (rapid-onset Cushing) ± androgens (virilization).

BOARD TRAP

A large (>4–6 cm), heterogeneous adrenal mass with high unenhanced CT density (>10 HU) and slow washout = suspicious for ACC, not a benign adenoma — go to surgery, don't just observe.

2. Open adrenalectomy for ACC

WHAT YOU SEE

'OPEN ADRENALECTOMY SUSPECTED ACC', surgeon at table

WHAT IT ENCODES

Suspected ACC is removed by OPEN adrenalectomy performed by an experienced surgeon, with en-bloc resection of the tumor and any locally invaded structures. The open approach protects the capsule from rupture and prevents peritoneal tumor seeding that would convert a curable tumor into incurable carcinomatosis.

BOARD TRAP

Even for a large mass, the open approach is chosen specifically to keep the capsule intact — capsule violation worsens prognosis.

3. Laparoscopic = X for ACC

WHAT YOU SEE

Red X over 'LAPAROSCOPIC TABLE'

WHAT IT ENCODES

Laparoscopic adrenalectomy is appropriate for small benign tumors but is CONTRAINDICATED when ACC is suspected because of the high risk of capsule rupture, tumor spillage and port-site/peritoneal recurrence.

BOARD TRAP

Choosing laparoscopic resection for a suspected ACC is the classic wrong answer — it risks spillage and recurrence; do an open procedure.

4. En-bloc + negative margins

WHAT YOU SEE

'ONLY CURE COMPLETE RESECTION NEGATIVE MARGINS', en-bloc

WHAT IT ENCODES

Cure depends on a complete EN-BLOC R0 resection with NEGATIVE margins, including resection of locally invaded adjacent organs and regional lymph node dissection. Incomplete (R1/R2) resection or capsular rupture markedly raises recurrence risk and is an indication for adjuvant mitotane.

BOARD TRAP

Positive margins or intraoperative tumor spillage convert a 'curative' operation into high-risk disease requiring adjuvant therapy.

5. Post-op steroid replacement

WHAT YOU SEE

Surgeon portrait 'RELATED POST-OP CORTISOL NEEDS'

WHAT IT ENCODES

A cortisol-secreting ACC chronically suppresses ACTH and the contralateral adrenal; after resection the patient is acutely adrenally INSUFFICIENT and needs glucocorticoid replacement (with stress-dosing perioperatively), tapered slowly over months as the HPA axis recovers.

BOARD TRAP

Post-resection fatigue, hypotension, nausea and hypoglycemia = expected adrenal insufficiency from a previously cortisol-secreting tumor — give steroids, don't miss it.

6. Mitotane potion

WHAT YOU SEE

Glowing 'MITOTANE POTION' bottle

WHAT IT ENCODES

Mitotane (an o,p'-DDD/insecticide derivative) is an ADRENOLYTIC drug that is selectively toxic to adrenocortical cells and inhibits steroidogenesis; it is the mainstay of medical and adjuvant therapy for ACC. Because it destroys cortical cells, it itself CAUSES adrenal insufficiency that must be covered with glucocorticoid replacement.

BOARD TRAP

Mitotane creates iatrogenic adrenal insufficiency — every patient on it needs concurrent glucocorticoid replacement.

7. Mitotane therapeutic level

WHAT YOU SEE

Thermometer 'MITOTANE LEVEL — 14 to 20 mg/L therapeutic, >20 neurotoxic, <14 no effect'

WHAT IT ENCODES

Mitotane has a NARROW therapeutic window requiring drug-level monitoring: target 14–20 mg/L. Below 14 mg/L it is ineffective; above 20 mg/L it causes dose-limiting NEUROTOXICITY (ataxia, confusion, lethargy). GI upset is the most common early side effect.

BOARD TRAP

Both under- and over-dosing are tested: <14 = no antitumor effect, >20 = neurotoxic — dose to the level, not by symptoms alone.

8. CYP3A4 induction → double hydrocortisone

WHAT YOU SEE

'CYP3A4 FURNACE' + 'GLUCOCORTICOID REPLACEMENT'

WHAT IT ENCODES

Mitotane is a potent CYP3A4 INDUCER and raises cortisol-binding globulin, so it accelerates hydrocortisone metabolism on top of causing adrenal insufficiency — patients typically need roughly DOUBLE the usual hydrocortisone replacement dose. The CYP3A4 induction also lowers levels of many co-administered drugs (e.g., warfarin, some chemo).

BOARD TRAP

A standard hydrocortisone dose is insufficient on mitotane — failure to double it (and account for CYP3A4 drug interactions) leads to under-replacement and crisis.

9. Mitotane indications

WHAT YOU SEE

'INDICATIONS MITOTANE — stage 3 or 4, Ki-67>10%, incomplete resection'

WHAT IT ENCODES

Adjuvant/therapeutic mitotane is reserved for HIGH-RISK or advanced ACC: stage III/IV disease, high proliferation (Ki-67 >10%), incomplete (R1/R2) resection or capsular rupture, and recurrent disease. Low-risk completely resected tumors may just be surveilled.

BOARD TRAP

Risk stratification (Ki-67, margins, stage) decides who gets adjuvant mitotane — not every resected ACC needs it.

10. EDP-M metastatic chemo

WHAT YOU SEE

'METASTATIC CHEMO STATION — EDP-M' (etoposide, doxorubicin, cisplatin + mitotane)

WHAT IT ENCODES

For advanced/metastatic ACC, first-line systemic therapy is the EDP-M regimen: Etoposide + Doxorubicin + cisPlatin combined with Mitotane (the FIRM-ACT trial standard). Lung is the most common site of metastasis.

BOARD TRAP

EDP-M is the go-to combination for metastatic ACC — mitotane is continued alongside the cytotoxic backbone.

11. Surveillance every 3-6 months

WHAT YOU SEE

'FOLLOW-UP CALENDAR — imaging every 3 to 6 months first 2 years'

WHAT IT ENCODES

ACC recurs early and often, so surveillance is intensive: cross-sectional imaging (and any tumor markers/hormones the tumor secreted) every 3–6 months for at least the first 2 years, then less frequently up to ~5–10 years. Most recurrences appear within the first 2 years.

BOARD TRAP

Because recurrence is early, surveillance intervals are short (3–6 mo) initially — don't space them out too soon.

12. Clinical pearl

WHAT YOU SEE

'OPEN ADRENALECTOMY FOR ACC. MITOTANE 14-20 THERAPEUTIC. CYP3A4 INDUCTION DOUBLE HYDROCORTISONE. FATIGUE+HYPOTENSION = ADRENAL INSUFFICIENCY. FOLLOW-UP 3-6 MO'

WHAT IT ENCODES

One-line synthesis: OPEN en-bloc R0 adrenalectomy is the only cure for ACC; mitotane (target 14–20 mg/L) is the adrenolytic adjuvant/medical therapy and EDP-M is added for metastatic disease; mitotane's CYP3A4 induction + adrenolysis means DOUBLE the hydrocortisone; post-op or on-mitotane fatigue/hypotension = adrenal insufficiency; surveil every 3–6 months.

BOARD TRAP

The high-yield triad: don't go laparoscopic for ACC, don't under-replace steroids on mitotane, and don't wait to treat the resulting adrenal insufficiency.